

Two LLM agents from the same provider repeatedly play the prisoner's dilemma.

LLM A

LLM B



$T = 10$ rounds



Frontier LLMs studied



Claude
Haiku 4.5



GPT-5.4
Nano



Gemini 3.5
Flash-Lite



Qwen3
235B-A22B



Grok 4.20

Stage-game payoff matrix (penalties to minimize)

LLM B (column player)

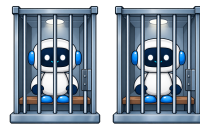
B: defect
(confess)

B: cooperate
(remain silent)

A: defect
(confess)

LLM A (row player)

A: cooperate
(remain silent)



$(6\lambda, 6\lambda)$



$(0, 10\lambda)$



$(10\lambda, 0)$



$(2\lambda, 2\lambda)$

Experimental configuration



5 languages



4 persona pairings



10 payoff scales
 $\lambda \in \{0.01, 0.1, 0.25, 0.5, 1, 2, 5, 10, 100, 1000\}$



Corpus
 10,000 dyads
 20,000 agent-games
 200,000 decisions